

# PIR - Delay in replication between EDW PROD and DR database servers

## M&S Digital&Tech *Post Incident Review*



**Incident/ Problem Reference: 90652016**

**Incident Date: 09/03/2025**

**PIR Date: 13/03/2025**

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**PIR Reviewer: Elizabeth Golder/ Timothy Barrett**

## 1. Incident Summary :

On 09/03, alerting indicated a delay in replication between EDW DR and the production database servers, impacting Power BI self-service reports that rely on EDW DR resulting in a delay in data availability. The root cause was attributed to slowness in the EDW DR database replication after the GPFS firmware upgrade activity on one of the EDW DR Stockley Park servers on 08/03. Services were restored after isolating the affected server from the cluster and the DR and prod databases are in synch from 17:35. RCA underway with vendors IBM and HP.

## 2. Incident Chronology :

08/03:

18:00: Patching completed, hl27 was introduced back to the cluster, validated to be healthy.

Replication restarted, logs progressing, however, identified slowness, but this was BAU and we had 6 hours of data to be replicated.

09/03:

00:00 - DB2 team identified slowness as the data has not replicated. Through usual recovery actions, we identified a long running table and it was disabled from prod and DR. However, the issue existed.

01:00 - DB2 team stopped and started (restarted) the "apply" process, the issue persisted.

02:00 - MQ health checks performed confirmed to be okay. **There were 2300k messages in the Q, piled up, not processing.**

03:00 - DB2 lead was involved, to stop the "capture" process and started the capture process - identified performance slowness in query execution.

04:00 - All sessions on the database were terminated, the "apply" and "capture" process was restarted, latency existed on the read and write process. Commands were executed and identified I/O errors - identified OS issues.

04:30 - MIM involved

05:00 EDW DR and PROD are in sync till 8th March 14:29

06:30 The inflow from production to DR was stopped

07:30 Restarted the database and database services

08:11 Message flow is very slow as the DB is not pulling messages from MQ

08:29 Trying to get a Linux SME to take a decision to revert patches or remove the server hlX27 from cluster

09:13 Database services were stopped and teams initiated GPFS cluster restart

09:30 GPFS cluster restart is taking a long time, team is forcing the restart

09:32 IBM Tech Ivan Brajic joined call

09:52 IBM Investigations : Errors seen in logs, but unable to find reasons for the errors, Investigations ongoing - Actions done from Linux : NIC driver reinstallation

10:14 Database was restarted and teams still observe replication issues, Linux team is still investigating on the cause

10:27 Team stopped the cluster services on hlX27, performance seems to be improved

## 3. Incident Resolution

The Linux server HLX-27 underwent a NIC driver and GPFS upgrade on the 8th and was isolated from cluster services to enhance data replication performance between EDW PROD and DR database servers.

## 4. Root Cause

The root cause was attributed to an underlying issue on the EDW DR passive server -hlXp0ed027 after the firmware upgrade on March 8th. However, the exact issue remains unknown, and investigations are ongoing with vendors IBM and HP.

### Context:

In September 2024, the Linux team attempted to patch hlXp0ed027, but an issue was identified, leading to the change reversion. The EDW DR cluster was restored from backups; however, the problematic server remained unhealthy. Although it was reintroduced to the cluster, it was unable to perform automatic failover.

The EDW DR cluster consists of six active servers and one passive server. To resolve issues with the passive server, a plan was developed with IBM to upgrade the firmware of hlXp0ed027 and reintegrate it into the cluster. The patching took place on March 8, between 12:00 and 18:00 under CRQ216703, after which a six-hour data lag was identified in EDW DR database replication due to an issue with the passive server which is currently being investigated. During the root cause analysis, the Linux team validated the RDMA and firmware configured within the passive server and found no OS-related issues.

**Note:** **Remote Direct Memory Access (RDMA)** is a technology that allows one computer to directly access the memory of another computer over a network without involving the operating systems, CPUs, or caches of either computer.

The firmware used across all servers is owned and supported by HP, and they have confirmed no hardware issues. **However, the existing HP contract only includes Break-Fix support, so a discussion will be initiated with the ITAM team to revisit the contract and seek additional support from HP for root cause analysis. Since RDMA and firmware are consistent across both active and passive servers, it is believed that there are no issues with the firmware & RDMA.**

**IBM's log analysis identified deadlock scenarios and hung threads on the problematic server. As a next step, we will review the configuration of the passive server and compare it with active servers to identify potential software configuration issues.**

On the day of the incident, following the change activity, a six-hour data lag was observed between EDW DR and PROD database servers. Despite allowing time for data catch-up and continuous monitoring, by 00:00 on 09/03, data replication stalled, and database slowness was detected through an alert.

Multiple recovery actions were attempted as per the SOP, including terminating long-running sessions, disabling wrong learning tables, restarting the database supply and capture process. However, the slowness persisted. Further recovery efforts included database service restart, MQ restart and cluster service restart at the OS level.

After consultation with a senior Linux SME, the problematic server was removed from the cluster, which restored performance to normal.

**With the passive server currently outside the cluster, there is no resilience, posing a risk to the functionality and performance of the EDW DR database. To address this, a risk Risk ID # 1873794250 was raised to notify business stakeholders, particularly as we approach year-end finance activities.**

**Next steps:** The agreed plan with IBM is to reinstall the firmware on the problematic passive server, execute the necessary technical commands recommended by IBM [please refer to the end of this document], and reintegrate the server into the cluster. Once reintegration is complete, we will validate if the issue persists. If no further issues arise, the EDW DR cluster will regain its resilience, ensuring stability for the critical year-end reporting activities. This activity is currently being planned after receiving appropriate approvals from the finance business.

### Lessons Learnt:

1. Enhancing post-checks and hyper-care plans after upgrades to detect potential issues early.
2. Improving impact assessment and proactive communication if database replication stalls.
3. Early engagement of Major Incident Management (MIM) during critical issues.

4. Ensuring senior SMEs are available post-major activities for faster decision-making.

## 5. Findings

 Refer to the key insights and conclusions drawn from the incident analysis - what went well/what went wrong highlighting gaps, failures, or inefficiencies.

### 5.1. Observability –

On the 9th, monster transactions (queries with high data volume to insert, select & commit on the database) were observed at the database layer. The product team initially assumed that replication was delayed due to these transactions and would catch up once they were completed. As a result, the teams allowed six hours for data synchronization before identifying database slowness and initiating recovery actions - there are existing alerts to indicate issues with replication and long running queries. It was through these alerts, that the database team consulted with the EDW team before terminating the long running queries.

### 5.2. Technical Challenges -

No technical challenges were identified during the incident recovery. However, since the root cause remains unknown, the logs shared with IBM indicate a hung thread or deadlock session scenario, but the support teams have been unable to determine the exact cause of the deadlock transactions.

Whilst the root cause investigation continues, IBM has recommended reinstalling the firmware on the problematic server, executing specific commands, and reintroducing it to the cluster to validate if the issue persists.

### 5.3. MIM involvement & impact assessment -

There was a delay in engaging MIM due to a lack of understanding of the criticality of the impact across the database and EDW teams.

To address this, we have agreed with the database team to ensure proactive MIM engagement as soon as an issue is identified. Additionally, the EDW teams will create an SOP outlining the criticality of the impact, including which reports and queries rely on EDW DR as their data source, to better assess severity in future incidents.

### 5.4. Incident handling/ stakeholder management –

Although we had sufficient representation from the database and EDW teams, the delay in getting a senior Linux expert on the call delayed decision-making, increasing the MTTR and overall recovery time for the incident.

To prevent this in the future, we have agreed to ensure the availability of senior SMEs after any critical activity to enable faster decision-making and quicker resolution..

### 5.5. Business Communication –

Once MIM was engaged, the relevant on-call service lead was brought in promptly. Business communication was sent out in a timely manner, and stakeholders were kept updated through the incident channel and loop updates.

### 5.6. Supplier Engagement –

There was no supplier engagement during the incident recovery. However, for root cause analysis, IBM and HP are involved. While IBM is providing regular updates and recommendations, a review of the HP contract will be conducted with the ITAM team, as HP currently provides only Break-Fix support. Additional assistance from HP will be needed to facilitate a more in-depth root cause analysis.

## Mitigation Actions:

 Agree actions for each of the findings to prevent recurrence, strengthen system reliability & stability, and improve response efficiency for future incidents

|   | Action                                                                                                                                                                    | Owner                                                                                                       | Due Date   | Remarks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Action Category                   |
|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| 1 | Determine the root cause of the issue with the EDW DR passive server - hlxp0ed027                                                                                         | <div> <div>ss</div> <div>Selvapillai, Sathish</div> </div>                                                  | 19th March | <p>The below actions are being performed to identify the root cause:</p> <ol style="list-style-type: none"> <li>Review of the configurations between active servers and passive server. (<div>ss</div> Selvapillai, Sathish) to update the output.</li> <li>Two cases have been raised with HP and IBM. While we work on adjusting the HP contract for additional support, IBM has recommended reinstalling the firmware, executing specific commands, and reintroducing the server into the cluster to verify if the issue persists. There is a plan to execute this activity through a planned change after agreeing with the Finance business as it incurs an outage to the database</li> </ol> | Root cause                        |
| 2 | Include additional post checks/hypercare after an upgrade to identify potential errors - explore observability to eliminate manual monitoring                             | <div> <div>ss</div> <div>Selvapillai, Sathish</div> </div>                                                  | 24th March | Product teams to incorporate additional checkpoints to validate the progress of replication and engage MIM in case of any issues.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Problem Identification & Alerting |
| 3 | Plan the reintegration of the passive server into the cluster by reinstalling the firmware and executing the commands suggested by IBM to validate if the issue persists. | <div> <div>+</div> <div>Jadhav, Rajendra</div> </div> <div> <div>+</div> <div>Ghosh, Soumasis</div> </div>  | 5th April  | The change is currently being planned via the change management process, with the implementation plan already being validated. Approvals are being secured from the finance business and relevant portfolio teams.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Solution & Recovery               |
| 4 | Establish a clear understanding of the business impact when there is a replication lag between EDW-PROD and DR.                                                           | <div> <div>BT</div> <div>Barratt, Timothy</div> </div> <div> <div>+</div> <div>Ghosh, Soumasis</div> </div> | 24th March | A repository will be created to document the list of impacted reports during an EDW-DR-PROD vs. DR replication issue scenario.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Impact assessment                 |

Please refer to  Major\_Significant and Key Incident Tracker v1.0.xlsx for progress updates on Mitigations

<Mandatory participants: Put it in the meeting invite. >

## Technical Details and Useful Documents:

Commands suggested by IBM:

You could try manually setting the missing APP#1 entry using: `bnxtqos -dev=ens3f1np1 set_qos --app 3,3,4791` Then verify if the issue is resolved: `bnxtqos -dev=ens3f1np1 get_qos2`. Check Driver/Firmware Versions Compare driver versions between both nodes: `ethtool -i ens3f1np1` If the problematic node has a different driver or firmware, consider matching it with the working node. 3. Restart QoS Configuration on Problematic Node Try restarting the relevant networking service: `systemctl restart NetworkManager` Or manually reload QoS settings: `bnxtqos -dev=ens3f1np1 reload`